

Practical 3

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Electricity and Electronics EBN 111

21 – 24 June 2021

Practical Information

Maximum marks:	50	Full marks:	50
Duration of practical:	180 min	Open/Closed book:	Open
Total number of pages (this page included)		13	

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Important

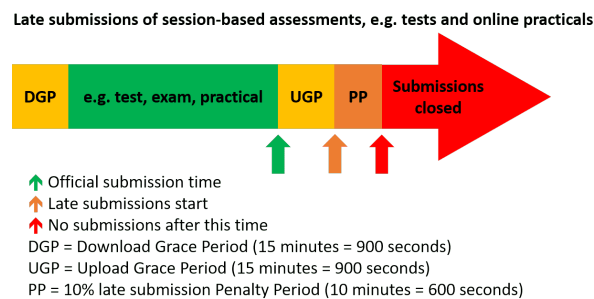
1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your practical, you declare the following:**

The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

Online assessment submission rules and time-line

The following rules and time-line apply for all practicals for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the practical for students to download all documents related to the practical and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the practical.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the practical.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.practicals@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final practical mark.
8. No late submissions are accepted after the PP.



Please Note:

TinkerCAD Circuits may round some numbers that are entered into the different instruments. As long as you make sure you enter the value exactly into the blue parameter box in TinkerCAD Circuits, the simulation will be correct, even if the instrument display shows a rounded value. This applies to all questions in the practical.

Questions 1 to 16 [20]

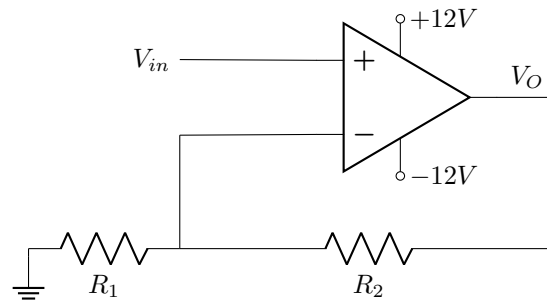
Overview:

In this question you will first solve an op-amp circuit theoretically. You will then build and test this op-amp circuit using TinkerCAD Circuits.

Instructions:

The circuit diagram in Figure 1 illustrates the op-amp circuit that you will build. Before building the circuit, answer the following theoretical question.

Figure 1:



Let the output voltage be $V_O = [V_O]$ and the input signal $V_{in} = [V_{in}]$. The values for resistors are $R_1 = [R_1]$ and $R_2 = [R_2]$. Perform theoretical analysis on the circuit in Figure 1 and set up the equation in the following form:

$$V_O = (A)V_{in} \quad (1)$$

What is the value of A ?

01 | A [2]

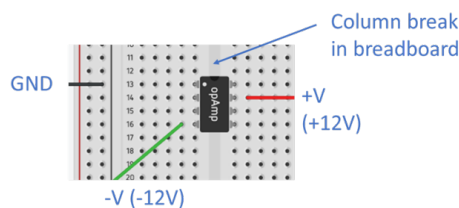
Next, you will implement this amplifier circuit using TinkerCAD Circuits by performing the following steps:

1. A 741 Operational Amplifier component (Figure 2) is available in TinkerCAD Circuits in the Integrated Circuits toolbox on the righthand side of the TinkerCAD Circuits workspace and will be used for building and testing a virtual op-amp circuit in this practical.

Figure 2:



2. The 741 operational amplifier can be placed on the breadboard so that pins 1 to 4 (left side) and 5 to 8 (right side) are separated by the column break on the breadboard, as shown in Figure 3. This also gives an indication as to how to connect power to the op-amp on the breadboard.

Figure 3:

3. Three power supplies will be required, as described in the Practical 3 guidelines document.
4. Set up two of the power supplies to provide positive (+12 V) and negative (-12 V) voltage supplies. This process is described in detail in the guidelines document. For the positive supply (+12 V), the power supply is set up as previously in Practical 2, with the negative port (black) connected to ground on the breadboard circuit and the positive port (red) connected to the positive voltage supply of the op-amp (+V, pin 7). For the negative supply (-12 V), the output ports of the power supply must be switched. This means that the positive port (red) must be connected to ground on the breadboard circuit, and the negative port (black) must be connected to the negative voltage supply of the op-amp (-V, pin 4).

Please Note: For the sake of convenience, power supply rails can be set up along the top and bottom of the breadboard. In order to establish a positive power supply rail at one end of the breadboard, note that the positive port (red output) of the power supply can be connected to the top rail or node on the breadboard labelled (+) while the ground (black output) can be connected to the top rail or node labelled (-). Conversely, a negative power supply at the other end of the breadboard may be established by connecting the positive port (red output) of the supply to the bottom rail or node on the breadboard labelled (-) while the ground (black output) can be connected to the bottom rail or node labelled (+). These can then be used as positive and negative voltage supply rails (+12 V and -12 V) and branches can be made from any points along these rails using wires to connect to the op-amp circuit. The grounds (-) of the power supplies must be connected on the breadboard circuit.

5. Set up the third power supply to provide the input voltage $V_{in} = [V_{in}]$, which will be connected at the non-inverting (+) input of the op-amp (pin 3). The same approach as in step 4 can be taken to create either a positive or negative supply. For a negative input voltage $V_{in} = [V_{in}]$, the positive port (red) must be connected to ground on the breadboard circuit, and the negative port (black) will be the input voltage $V_{in} = [V_{in}]$. You can double check to make sure that you have configured the power supply correctly to produce a positive or negative voltage by testing this with a multimeter, as detailed in the guidelines document.

Important: The grounds of all the power supplies must be connected on the breadboard circuit. This can be done using wires to connect all the grounds to a common ground node or rail.

6. As shown in Figure 1, proceed by connecting pin 6 of the op-amp to pin 2 of the op-amp using a resistor $R_2 = [R_2]$ (Note: You may use wires to extend the connection between nodes).
7. Connect pin 2 of the op-amp to ground (0 V) using a resistor $R_1 = [R_1]$.
8. Connect pin 4 of the op-amp to the -12 V rail of the power supply.
9. Connect pin 7 of the op-amp to the +12 V rail of the power supply.
10. Connect the third power supply – which provides the input voltage $V_{in} = [V_{in}]$ – to pin 3 of the op-amp. Remember: the grounds for all the power supplies should be connected to a common ground node or rail.

11. Connect pin 6 of the op-amp to the red port of a virtual multimeter. This is where the output will be displayed. Again, the black port of the multimeter should go to the common ground node or rail.
12. Set the input voltage $V_{in} = [V_{in}]$ (third power supply) and then read off the corresponding multimeter value for the measured output voltage V_O along with the corresponding unit.
13. Different values for the input voltage $V_{in} = [V_{in}]$ (third power supply) are provided, and in each case the voltage V_O should be determined.

What is the measured output voltage V_O ?

02 V_O if $V_{in} = [V_{in1a}]$ [1]

03 V_O if $V_{in} = [V_{in1b}]$ [1]

04 V_O if $V_{in} = [V_{in1c}]$ [1]

05 V_O if $V_{in} = [V_{in1d}]$ [1]

06 V_O if $V_{in} = [V_{in1e}]$ [1]

07 V_O if $V_{in} = [V_{in1f}]$ [1]

08 V_O if $V_{in} = [V_{in1g}]$ [1]

09 V_O if $V_{in} = [V_{in1h}]$ [1]

10 V_O if $V_{in} = [V_{in1i}]$ [1]

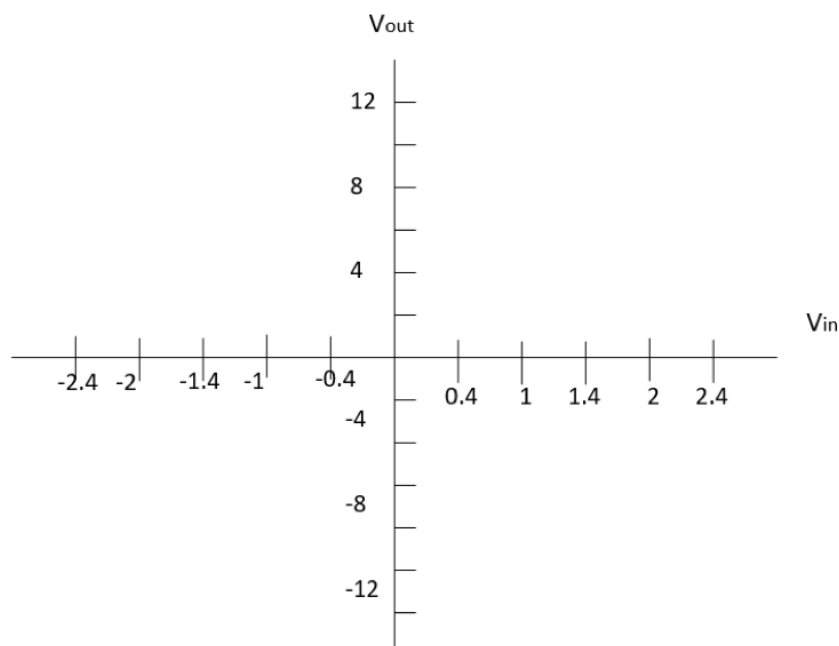
11 V_O if $V_{in} = [V_{in1j}]$ [1]

12 V_O if $V_{in} = [V_{in1k}]$ [1]

13 V_O if $V_{in} = [V_{in1l}]$ [1]

14. By using the values from Questions 2 to 13 one can draw the graph of V_O (V_{out}) versus V_{in} . Drawing the graph is not necessary but Figure 4 can assist in determining the gradient of the straight line of the op-amp.

Figure 4:



15. Calculate the gradient of the straight line (linear region) of the op-amp.	
What is the gradient G of the linear region?	
14	G [2]
16. Calculate the percentage difference between the measured gradient G (Question 14) and the theoretical gain A (Question 1) of the op-amp. $\%diff = \left \frac{A - G}{A} \right \times 100 \quad (2)$	
17. Complete the below questions	
What is the percentage difference between the measured gradient G and the theoretical gain A ?	
15	%diff [2]
True or false: the circuit is an example of a non-inverting amplifier.	
16	1 : True or 2 : False [2]
Please note: When selecting an option for Question 16, please enter either <u>1 for true</u> OR <u>2 for false</u> .	

Questions 17 to 32 [20]

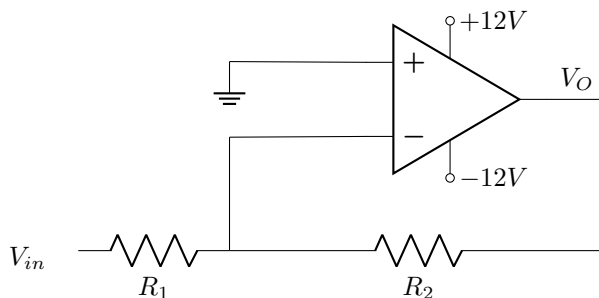
Overview:

In this question you will first solve an op-amp circuit theoretically. You will then build and test this op-amp circuit using TinkerCAD Circuits.

Instructions:

The circuit diagram in Figure 5 illustrates the op-amp circuit that you will build. Before building the circuit, answer the following theoretical question.

Figure 5:



Let the output voltage be $V_O = [V_O]$ and the input signal $V_{in} = [V_{in}]$. The values for resistors are $R_1 = [R_1]$ and $R_2 = [R_2]$. Perform theoretical analysis on the circuit in Figure 5 and set up the equation in the following form:

$$V_O = (A)V_{in} \quad (3)$$

What is the value of A ?

17 | A [2]

Next, you will implement this amplifier circuit using TinkerCAD Circuits by performing the following steps:

1. Locate a 741 Operational Amplifier component (Figure 2) in TinkerCAD Circuits from the Integrated Circuits toolbox on the righthand side of the TinkerCAD Circuits workspace. Place the 741 operational amplifier on the breadboard so that pins 1 to 4 (left side) and 5 to 8 (right side) are separated by the column break on the breadboard, as shown in Figure 3.
2. Three power supplies will be required, as described in the Practical 3 guidelines document.

3. Set up two of the power supplies to provide positive (+12 V) and negative (-12 V) voltage supplies. This process is described in detail in the guidelines document. For the positive supply (+12 V), the power supply is set up as previously in Practical 2, with the negative port (black) connected to ground on the breadboard circuit and the positive port (red) connected to the positive voltage supply of the op-amp (+V, pin 7). For the negative supply (-12 V), the output ports of the power supply must be switched. This means that the positive port (red) must be connected to ground on the breadboard circuit, and the negative port (black) must be connected to the negative voltage supply of the op-amp (-V, pin 4).

Please Note: For the sake of convenience, power supply rails can be set up along the top and bottom of the breadboard. In order to establish a positive power supply rail at one end of the breadboard, note that the positive port (red output) of the power supply can be connected to the top rail or node on the breadboard labelled (+) while the ground (black output) can be connected to the top rail or node labelled (-). Conversely, a negative power supply may be established at the other end of the breadboard by connecting the positive port (red output) of the power supply to the bottom rail or node on the breadboard labelled (-) while the ground (black output) can be connected to the bottom rail or node labelled (+). These can then be used as the positive and negative voltage supply rails (+12 V and -12 V) and branches can be created from any points along these rails using wires to connect to the op-amp circuit. The grounds (-) of the power supplies must be connected on the breadboard circuit.

4. Set up the third power supply to provide the input voltage $V_{in} = [V_{in}]$, which will be connected to resistor $R_1 = [R_1]$. The same approach as in step 3 can be taken to create either a positive or negative supply. For a negative input voltage $V_{in} = [V_{in}]$, the positive port (red) must be connected to ground on the breadboard circuit, and the negative port (black) will be the input voltage $V_{in} = [V_{in}]$. You can double check to make sure that you have configured the power supply correctly to produce a positive or negative voltage by testing this with a multimeter, as detailed in the guidelines document.

Important: The grounds of all the power supplies must be connected on the breadboard circuit. This can be done using wires to connect all the grounds to a common ground node or rail.

5. As shown in Figure 5, proceed by connecting pin 6 of the op-amp to pin 2 of the op-amp using a resistor $R_2 = [R_2]$ (Note: You may use wires to extend the connection between nodes).
6. Connect pin 3 of the op-amp to ground (0 V).
7. Connect pin 4 of the op-amp to the -12 V rail of the power supply.
8. Connect pin 7 of the op-amp to the +12 V rail of the power supply.
9. Connect the third power supply – which provides the input voltage $V_{in} = [V_{in}]$ – to pin 2 of the op-amp using resistor $R_1 = [R_1]$. Remember: the grounds for all the power supplies should be connected to a common ground node or rail.
10. Connect pin 6 of the op-amp to the red port of a virtual multimeter. This is where the output will be displayed. Again, the black port of the multimeter should go to the common ground node or rail.

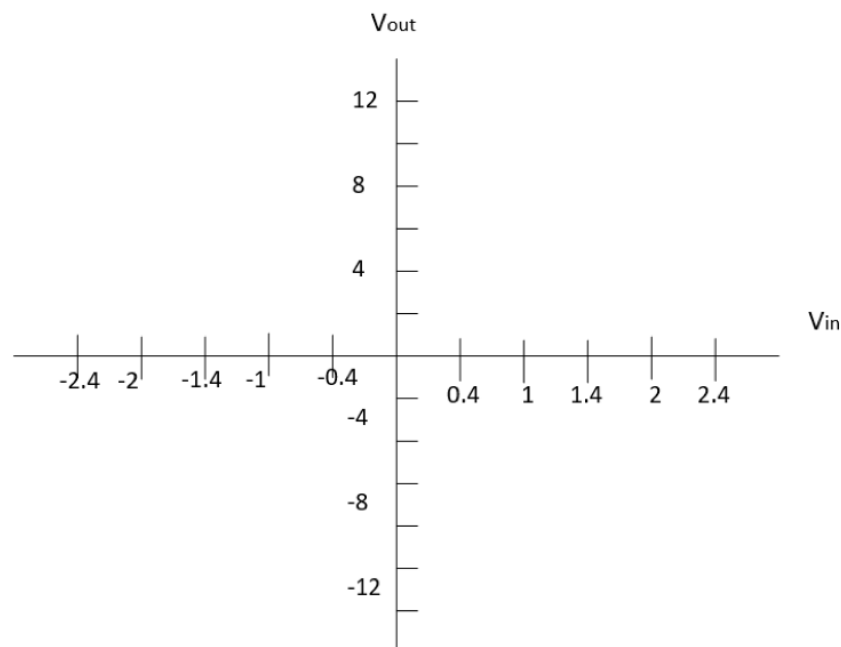
11. Different values for the input voltage $V_{in} = [V_{in}]$ (third power supply) are provided, and in each case the voltage V_O should be determined.

What is the measured output voltage V_O ?

18	V_O if $V_{in} = [V_{in2a}]$ [1]
19	V_O if $V_{in} = [V_{in2b}]$ [1]
20	V_O if $V_{in} = [V_{in2c}]$ [1]
21	V_O if $V_{in} = [V_{in2d}]$ [1]
22	V_O if $V_{in} = [V_{in2e}]$ [1]
23	V_O if $V_{in} = [V_{in2f}]$ [1]
24	V_O if $V_{in} = [V_{in2g}]$ [1]
25	V_O if $V_{in} = [V_{in2h}]$ [1]
26	V_O if $V_{in} = [V_{in2i}]$ [1]
27	V_O if $V_{in} = [V_{in2j}]$ [1]
28	V_O if $V_{in} = [V_{in2k}]$ [1]
29	V_O if $V_{in} = [V_{in2l}]$ [1]

12. Use the values from Questions 15 to 25 to draw the graph of V_{out} (V_O) versus V_{in} (as shown in Figure 6). Drawing the graph is not necessary but Figure 6 can assist in determining the gradient of the straight line of the op-amp.

Figure 6:

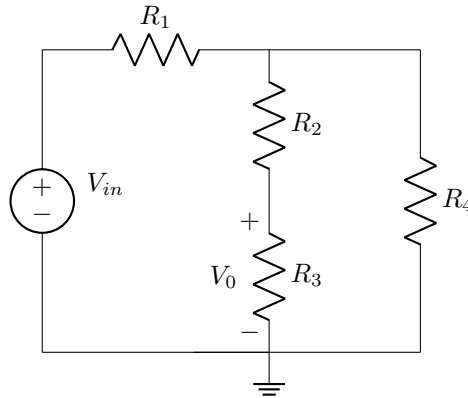


13. Calculate the gradient of the straight line (linear region) of the op-amp.	
What is the gradient G of the linear region?	
30	G [2]
14. Calculate the percentage difference between the measured gradient G (Question 17) and the theoretical gain A (Question 30) of the op-amp.	
$\%diff = \left \frac{A - G}{A} \right \times 100 \quad (4)$	
15. Complete the below questions	
What is the percentage difference between the measured gradient G and the theoretical gain A ?	
31	$\%diff$ [2]
True or false: the circuit is an example of a non-inverting amplifier.	
32	1 : True or 2 : False [2]
Please note: When selecting an option for Question 28, please enter either <u>1 for true</u> OR <u>2 for false</u> .	

Questions 33 to 34 [4]

Overview:

In this question you will build and test a circuit with an AC input using TinkerCAD Circuits.

Instructions:**Figure 7:**

1. Build the circuit in Figure 7 in TinkerCAD Circuits. The values for the resistors are $R_1 = [R_1]$, $R_2 = [R_2]$, $R_3 = [R_3]$ and $R_4 = [R_4]$.
2. Use the function generator to generate a sine wave with frequency $f = [f]$, peak-to-peak voltage $V_{pp} = [V_{pp}]$, and offset voltage given by [Offset]. V_{in} is the signal that will be generated by the function generator after setting it up with the given parameters.
Please Note: Peak-to-peak voltage is the voltage difference between the maximum and minimum voltage of a signal.
3. Connect the oscilloscope across the resistor $R_3 = [R_3]$ to measure V_0 as indicated in Figure 7. Set the time per division setting of the oscilloscope $t/div = [t/div]$
4. Run the simulation and wait until the measured signal on the oscilloscope has stabilised. Observe and measure the voltage V_0 as displayed on the oscilloscope.

Please Note: Frequency is the inverse of the period.

What is the peak-to-peak voltage V_0 across R_3 ?

33 | V_{0pp} [2]

What is the frequency of the voltage V_0 across R_3 ?

34 | f [2]

Questions 35 to 37 [6]

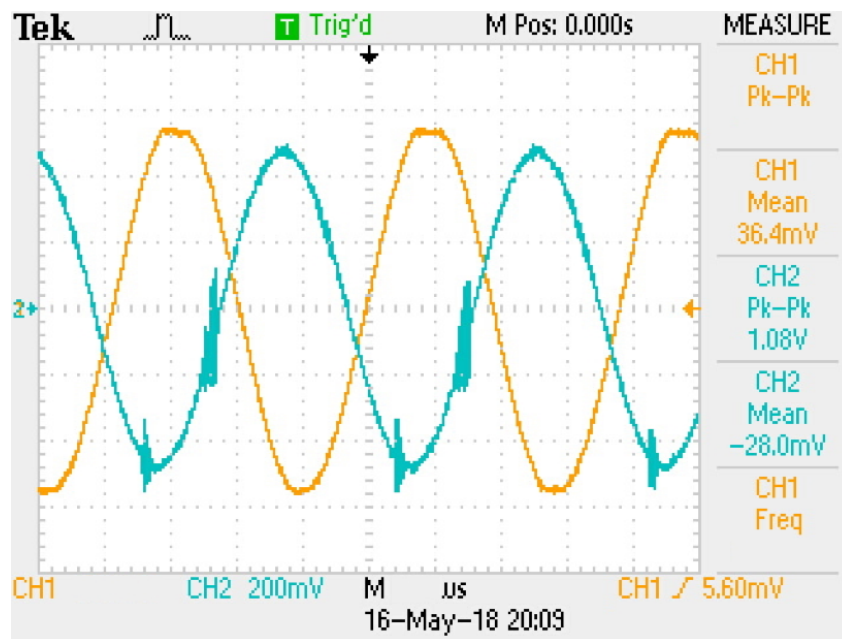
Overview:

In this question you will take measurements from a real oscilloscope screenshot.

Instructions:

Two voltages in a circuit were measured using a real oscilloscope. A screenshot of the measurement is shown in Figure 8 where the yellow signal is channel 1 (CH1) and the blue signal is channel 2 (CH2).

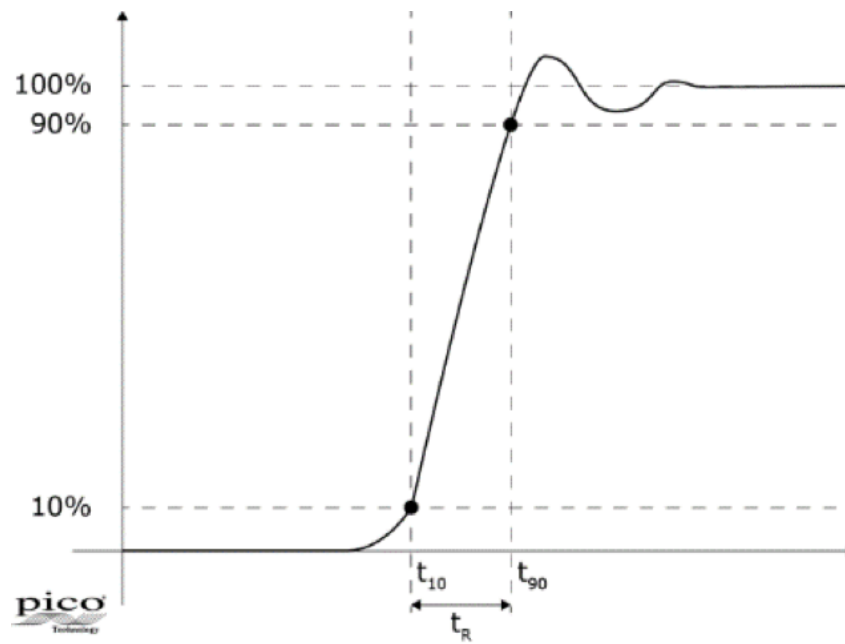
Figure 8:



1. Use the following values for Figure 8 in your calculations: time per division $t/div = [t/div]$ and voltage per division $V/div = [V/div]$.

Please Note: Divisions are defined by the dotted lines in Figure 8.

Please Note: Rise time, t_R , is the time it takes a signal to transition from 10% of its peak-to-peak voltage at t_{10} , to 90% of its peak-to-peak voltage at t_{90} . An example of measuring rise time is shown in Figure 9.

Figure 9:

<https://www.picotech.com/library/oscilloscopes/rise-time>

What is the peak-to-peak voltage V_{pp} measured on CH1?

35 | V_{pp} [2]

What is the frequency f of the signal measured on CH1?

36 | f [2]

What is the rise time t_R of the signal measured on CH1?

37 | t_R [2]

Grand Total : [50]